NETVISOR Mini Discovery V4

Project Information

Project Name: NETVISOR
Module: Discovery Engine
Version: V4
Author: Moad
Date: 2025

---

1. Overview

NETVISOR Mini Discovery V4 represents a major step in the evolution of the network discovery system.

After V1 (Ping Discovery), V2 (ARP + MAC Discovery), and V3 (TCP Service Detection), V4 introduces **Asset Identification Layer**, allowing the system to enrich discovered devices with human-readable identity information.

This includes:

- Hostname resolution
- Vendor identification (MAC-based OUI mapping)

V4 transforms NETVISOR from a technical scanner into a **real asset identification engine**.

---

2. Objectives

The main objectives of V4 are:

- Maintain Ping-based discovery (L3 reachability)
- Maintain ARP-based MAC discovery (L2 mapping)
- Maintain TCP port scanning (service detection)
- Add hostname resolution (reverse DNS)
- Add vendor identification using MAC prefix (OUI)

- Improve asset readability and classification
- Prepare for database integration (V5)

---

3. Architecture

Directory Structure

netvisor-mini-discovery-v4/

core/

- Network.php
- Scanner.php

modules/

- Ping.php
- Arp.php
- Tcp.php
- Hostname.php
- Vendor.php

services/

- DeviceBuilder.php

models/

- Device.php

run.php

---

4. Components

4.1 Ping Module

Responsible for:

- Detecting active hosts using ICMP
- Filtering reachable IPs

4.2 ARP Module

Responsible for:

- Extracting IP → MAC mappings
- Using system ARP table (ip neigh)
- Providing Layer 2 visibility

4.3 TCP Module

Responsible for:

- Port scanning (22, 80, 443)
- Detecting open services
- Basic service enumeration

4.4 Hostname Module

New in V4.

Responsible for:

- Reverse DNS lookup
- Mapping IP → hostname
- Providing human-readable device identity

Function:

- resolveHostname(ip)

4.5 Vendor Module

New in V4.

Responsible for:

- Identifying device manufacturer
- Using MAC prefix (OUI mapping)
- Basic vendor classification

Example vendors:

- TP-Link
- Huawei
- Dell

---

5. Discovery Workflow

Phase 1: Network Detection

Identify local subnet.

Example:

192.168.1.0/24

Phase 2: Ping Sweep

Detect active hosts.

Output:

List of reachable IPs.

Phase 3: ARP Collection

Retrieve MAC addresses from system ARP table.

Phase 4: TCP Scanning

Check open ports:

- 22 (SSH)
- 80 (HTTP)
- 443 (HTTPS)

Phase 5: Hostname Resolution

Perform reverse DNS lookup:

IP → Hostname

Phase 6: Vendor Identification

Map MAC prefix to vendor name.

Phase 7: Asset Enrichment

Combine all collected data into structured device profiles.

---

6. Features Implemented

Implemented in V4:

✓ Ping-based discovery
✓ ARP-based MAC discovery
✓ TCP port scanning
✓ Hostname resolution
✓ Vendor identification
✓ Asset enrichment
✓ Modular architecture
✓ CLI execution

7. Example Output

IP: 192.168.1.105
HOSTNAME: desktop-it
VENDOR: Dell
MAC: 0e:24:2c:a3:3e:23
PORTS: 80, 443

IP: 192.168.1.254
HOSTNAME: router.local
VENDOR: Huawei
MAC: e0:3d:a6:95:4b:50
PORTS: 80

TOTAL DEVICES: 8

8. Improvements Over V3

Feature	V3	V4
Ping Discovery	✓	✓
ARP Discovery	✓	✓
TCP Scanning	✓	✓
Hostname Resolution	✗	✓
Vendor Detection	✗	✓
Asset Readability	Medium	High
Device Intelligence	Low	Improved

9. Current Limitations

Not implemented yet:

✗ Database storage
✗ Multi-network discovery
✗ Topology mapping
✗ Router detection
✗ Service banner grabbing
✗ OS fingerprinting
✗ Real-time monitoring engine

---

10. Technical Challenges

During V4 development, several issues were encountered:

- Function name conflict with PHP built-in functions (gethostname)
- Inconsistent reverse DNS results
- Missing ARP cache entries for inactive devices
- Vendor mapping limited to small OUI dataset
- Some devices blocking ICMP or TCP probes

```
=======================================
||||||NETVISOR MINI DISCOVERY V4||||||
=======================================

Phase 1: Ping Sweep...
Phase 2: ARP Table...
===== DEVICES FOUND =====

IP: 192.168.1.1
HOSTNAME: gpon.net
VENDOR: Unknown Vendor
MAC: 28:77:77:99:a1:0c
PORTS: 80, 443

-------------------

IP: 192.168.1.5
HOSTNAME: unknown
VENDOR: Unknown Vendor
MAC: unknown
PORTS: none

-------------------

IP: 192.168.1.9
HOSTNAME: unknown
VENDOR: Unknown Vendor
MAC: unknown
PORTS: none

-------------------

IP: 192.168.1.32
HOSTNAME: unknown
VENDOR: Unknown Vendor
MAC: unknown
PORTS: none

-------------------
```

11. Future Roadmap

Version 5

Database Layer:

- Store discovered devices
- Track historical changes
- Detect new/removed devices

Version 6

Multi-Network Discovery:

- Multiple subnets support
- Network registry system

Version 7

Topology Engine:

- Router detection
- Network graph generation
- Relationship mapping

Version 8

Monitoring Engine:

- Metrics collection
- Health checks
- Alert system

---

12. Conclusion

NETVISOR Mini Discovery V4 significantly enhances the intelligence of the discovery engine by adding hostname resolution and vendor identification.

This version shifts NETVISOR from a simple network scanner into a structured asset identification system, laying the foundation for enterprise-level features such as database tracking, topology mapping, and real-time monitoring.